

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	PEP-8 style and consistent formatting followed.
2	Proper File Structure	Files and folders are logically organized	Yes	Project files are separated into templates, static files, and application code.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables are named according to their functionality.
4	Function / Method Names	Functions are descriptively named	Yes	Functions use descriptive names for prediction and preprocessing tasks.
5	Code Comments	Inline and block comments explain logic	Yes	Comments added for important logic and model implementation.
6	Modular Design	Code is split into reusable functions/modules	Yes	Reusable functions created for preprocessing and prediction.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused and duplicate code removed during development.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Model Training and Prediction Pipelines	High

2	Flood Prediction Model	Python, Scikit-learn, XGBoost	Flask Web Application and Model Evaluation	High
3	Flask Routes and Controllers	Flask, Python	User Input Processing and Prediction Dashboard	Medium
4	Input Validation Module	Python	Data Entry Forms and Prediction Requests	Medium
5	User Interface Components	HTML, CSS, JavaScript, Bootstrap	Prediction Dashboard and Result Pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with modular components.
Readability	5	Clear naming conventions and maintainable code.
Reusability	4	Several reusable modules for prediction and preprocessing.
Documentation / Comments	4	Comments provided for key functions and workflows.
Overall Score	5	Codebase is structured, maintainable, and suitable for future enhancements.